

Stacked Solar-Shade Radiator Architecture

Mission Scenario Analysis — Rev B

Scenario Analysis | Rev B — September 2026 | Not for Hardware Commitment

1. Summary

This analysis re-derives the February 2026 scenario matrix from the Rev B model, with the view factor computed for the real canopy geometry. The canopy earns its place in one regime: fixed assets in the equatorial belt that must run through lunar noon with dust-degraded radiators. Elsewhere an unshaded radiator sized to dusty end-of-mission properties performs as well or better and is simpler.

The Aegis Station Infrastructure surface program is south-polar, where the sun's elevation keeps a horizontal radiator's solar load small; the rover and the Short Hopper carry unshaded radiators and the roof interface plate. Polar power is a separate question, answered by vertical arrays, batteries, fission surface power and possibly transmission from lit high ground rather than by anything on a roof. The scenario analysis is offered as a reference for assets at low latitudes.

2. Scenario matrix

Scenario	Sun elevation	Design angle	Unshaded clean / dusty (W/m²)	Roof stack 0.4 m	Canopy 2 m (1 m overhang)	Verdict
South pole, nominal	2–6°	6°	444 / 357	74 / 56	167 / 137	Not recommended — unshaded radiator performs better
South pole, dust-degraded radiator	2–6°	6°	444 / 357	74 / 56	167 / 137	Not recommended — unshaded radiator performs better
Mid-latitude, dawn/dusk	5–20°	20°	405 / 276	74 / 56	167 / 137	Not recommended — unshaded radiator performs better
Mid-latitude, midday	30–60°	45°	345 / 152	74 / 56	167 / 137	Not recommended — unshaded radiator performs better
Equatorial, near noon	70–90°	80°	300 / 58	74 / 56	167 / 137	Justified for dust-degraded end of mission; marginal when clean
Mobile platform, any latitude	2–90°	45°	345 / 152	74 / 56	167 / 137	Not recommended — a 2 m canopy is not a vehicle roof; use an unshaded radiator and a tracking array
Static outpost, constrained footprint	30–90°	60°	319 / 98	74 / 56	167 / 137	Justified for dust-degraded end of mission; marginal when clean
Permanently shadowed region	—	—	461 / 393	74 / 56	167 / 137	Not applicable — no sun to shade; radiator sees sky only

Roof stack: same-size array 0.4 m above the radiator, the February configuration. Canopy: 2 m height with 1 m overhang all round. Both with coated backside at 330 K. The equatorial and constrained-footprint rows improve further with a 3–4 m canopy (Section 3).

3. Scenarios

1. South pole, nominal

Environment. Sun elevation 2–6°; sizing at 6°. **Unshaded radiator.** 444 W/m² clean, 357 W/m² dusty. **Shaded.** Roof stack 74 / 56; 2 m canopy 167 / 137 W/m²; overhang for full shade at the design angle 19.0 m per edge at 2 m. **Verdict.** Not recommended — unshaded radiator performs better.

2. South pole, dust-degraded radiator

Environment. Sun elevation 2–6°; sizing at 6°. **Unshaded radiator.** 444 W/m² clean, 357 W/m² dusty. **Shaded.** Roof stack 74 / 56; 2 m canopy 167 / 137 W/m²; overhang for full shade at the design angle 19.0 m per edge at 2 m. **Verdict.** Not recommended — unshaded radiator performs better.

3. Mid-latitude, dawn/dusk

Environment. Sun elevation 5–20°; sizing at 20°. **Unshaded radiator.** 405 W/m² clean, 276 W/m² dusty. **Shaded.** Roof stack 74 / 56; 2 m canopy 167 / 137 W/m²; overhang for full shade at the design angle 5.5 m per edge at 2 m. **Verdict.** Not recommended — unshaded radiator performs better.

4. Mid-latitude, midday

Environment. Sun elevation 30–60°; sizing at 45°. **Unshaded radiator.** 345 W/m² clean, 152 W/m² dusty. **Shaded.** Roof stack 74 / 56; 2 m canopy 167 / 137 W/m²; overhang for full shade at the design angle 2.0 m per edge at 2 m. **Verdict.** Not recommended — unshaded radiator performs better.

5. Equatorial, near noon

Environment. Sun elevation 70–90°; sizing at 80°. **Unshaded radiator.** 300 W/m² clean, 58 W/m² dusty. **Shaded.** Roof stack 74 / 56; 2 m canopy 167 / 137 W/m²; overhang for full shade at the design angle 0.4 m per edge at 2 m. **Verdict.** Justified for dust-degraded end of mission; marginal when clean.

6. Mobile platform, any latitude

Environment. Sun elevation 2–90°; sizing at 45°. **Unshaded radiator.** 345 W/m² clean, 152 W/m² dusty. **Shaded.** Roof stack 74 / 56; 2 m canopy 167 / 137 W/m²; overhang for full shade at the design angle 2.0 m per edge at 2 m. **Verdict.** Not recommended — a 2 m canopy is not a vehicle roof; use an unshaded radiator and a tracking array.

7. Static outpost, constrained footprint

Environment. Sun elevation 30–90°; sizing at 60°. **Unshaded radiator.** 319 W/m² clean, 98 W/m² dusty. **Shaded.** Roof stack 74 / 56; 2 m canopy 167 / 137 W/m²; overhang for full shade at the design angle 1.2 m per edge at 2 m. **Verdict.** Justified for dust-degraded end of mission; marginal when clean.

8. Permanently shadowed region

Environment. Sun elevation none; sizing at 0°. **Unshaded radiator.** 461 W/m² clean, 393 W/m² dusty. **Shaded.** Roof stack 74 / 56; 2 m canopy 167 / 137 W/m². **Verdict.** Not applicable — no sun to shade; radiator sees sky only.

4. Canopy height and latitude

For the fixed-asset cases, the decisive variables are canopy height and site latitude. Shaded values are clean / dusty at noon for a canopy sized to shade within 20° of the noon elevation.

Latitude	Noon elevation	Unshaded clean / dusty	Canopy 2 m	Canopy 3 m	Canopy 4 m	Verdict
0°	90°	297 / 52	205 / 170	251 / 210	280 / 235	Recommended, dust-driven
15°	75°	303 / 64	121 / 97	153 / 125	173 / 142	Recommended, dust-driven
30°	60°	319 / 98	47 / 32	65 / 48	76 / 58	Not recommended
45°	45°	345 / 152	-14 / -21	-7 / -15	-3 / -12	Not recommended
60°	30°	379 / 223	-56 / -58	-56 / -57	-55 / -57	Not recommended
85°	5°	447 / 363	-64 / -65	-64 / -65	-64 / -64	Not recommended

5. Program application

Aegis-Class Rover (polar, mobile). Unshaded 6 m² roof radiator, 2,143 W dusty; no roof array; tracking wings for power. The Rev A roof stack would have left it 445 W.

Luna–Aegis Short Hopper (polar). Roof interface plate only; vehicle thermal control is a separate SRR item.

LUNET nodes and the polar base. Unshaded radiators. Power at the pole comes from vertical arrays of the kind NASA is now soliciting, batteries for the shadow periods, fission surface power as it becomes available, and possibly transmission lines from persistently lit ridges; the canopy is not part of that mix.

Where it would apply. An equatorial ISRU demonstrator or habitat running through noon, a low-latitude logistics node, or a CLPS-class fixed payload with a dusty radiator and a power need. Offered as a reference for programs operating there.

Rev B, September 2026. Replaces the February 2026 scenario analysis.
